

Independent Replication Report — BVBRC-115

Silva et al. 2019 — *Bacillus velezensis* UFLA258 genome and clade comparison

Ollie / OpenClaw Wave-Keeper Replication (2026-07-05)

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Abstract

Silva et al. (*Genome Biology and Evolution* 2019, DOI 10.1093/gbe/evz208, PMID 31580420) report the complete genome of *Bacillus velezensis* strain UFLA258 (a Brazilian cotton-soil bio-control isolate) and compare it against every complete *B. velezensis*, *B. amyloliquefaciens*, and *B. siamensis* genome in GenBank as of Jun 2019 ($n=115$). Central claims: (i) UFLA258 is a 3.95 Mb, 46.69% GC single circular chromosome with 3949 CDS, 84 tRNA, 27 rRNA; (ii) Average Nucleotide Identity (ANI) $>95\%$ and digital DNA–DNA hybridization (dDDH) $>70\%$ cleanly separate the three species; (iii) rpoB β -subunit phylogeny confirms the species boundary; (iv) 19 GenBank strains deposited as *B. amyloliquefaciens* are actually *B. velezensis*, giving a final split of 105/9/1; (v) 12 biosynthetic gene clusters (BGCs) are detected in *B. velezensis*, of which 5 NRPS (bacilysin, bacillibactin, fengycin, bacillaene, surfactin) and 2 PKS (difficidin, macrolactin) are conserved core. This replication re-fetched the paper’s own deposited genome (CP039297) plus four canonical references (FZB42, UCMB5113, DSM7, *B. siamensis* SCSIO 05746), re-ran genome-stats, fastANI, prokka+MAFFT rpoB, and antiSMASH v8 with KnownClusterBlast on uicgpu, and independently confirmed every quantitative claim tested with $\leq 0.5\%$ deviation. Verdict: **REPLICATED (partial)** — claims C1–C7 fully reproduce; C8 verified on 1/19 reclassified strains (UCMB5113); C9 (full 115-genome PCA) and C10 (CRISPR ratio) descope for scale.

1 Paper summary

The paper delivers a Genome Announcement (of UFLA258) fused with a Genome-Biology-and-Evolution-scale comparative analysis. UFLA258 was Illumina-sequenced (NextSeq-500, $849\times$ coverage), assembled through the PATRIC auto pipeline (BayesHammer $\rightarrow$ Velvet / IDBA / SPAdes selected by QUAST $\rightarrow$ CONTIGuator scaffolding against *B. velezensis* UCMB5113 $\rightarrow$ FGAP + BlastN + CLC gap closure $\rightarrow$ circle), and annotated in RASTtk with Artemis manual curation. Comparative genomics used JspeciesWS (ANI), Kostas Lab (dDDH), MAFFT + MEGA10 (rpoB ML tree with T92+G+I, 1000 bootstraps), CRISPRfinderCAS, PHASTER, and antiSMASH 4.0.2. Taxonomic reclassifications flow from the AND of $\text{ANI} \geq 95\%$ and $\text{dDDH} \geq 70\%$.

2 Claims table

ID	Claim	Type	Tested?
C1	UFLA258 $\approx$ 3.95 Mb circular	quantitative	yes
C2	GC = 46.69%	quantitative	yes
C3	3 949 CDS, 27 rRNA, 84 tRNA	quantitative	yes
C4	ANI >95% delimits species	quantitative	yes
C5	rpoB UFLA258 vs FZB42/DSM7/siamensis = 99.7/98.9/98.5%	quantitative	yes

3 Method (numbered)

1. **Source text.** JATS-NXML pulled from PMC OAI-PMH; publisher PDF blocked by Cloudflare Turnstile at `academic.oup.com`, PMC PDF blocked by PoW at `pmc.ncbi.nlm.nih.gov`. `extraction/marker.md` and `extraction/nougat.mmd` are the JATS-derived plaintext with provenance headers. `paper.pdf` is a pandoc + xelatex render of the derived Markdown.
2. **Genomes** pulled via NCBI E-utilities `efetch` (no auth, free): UFLA258 (CP039297.1), FZB42 (NC_009725.2), UCMB5113 (HG328254.1), DSM7 (FN597644.1), *B. siamensis* SC-SIO 05746 (NZ_CP025001.1).
3. **Genome stats.** Length + GC from FASTA (BioPython); CDS/tRNA/rRNA/pseudo counts from the deposited GBK feature table for UFLA258.
4. **Pairwise ANI.** `fastANI v1.34 -fragLen 3000 -threads 16` over all 5×5 pairs.
5. **rpoB extraction + %ID.** Prokka v1.12 (`-fast`) annotated each genome; the "DNA-directed RNA polymerase subunit beta" CDS (3 582 nt in all 5) pulled by exact-string match, MAFFT v7.526 (`-auto`) aligned, pairwise %ID over ungapped columns.
6. **BGC profiling.** antiSMASH v8.0.4, first default, then rerun with `-cb-knownclusters` for MIBiG-KCB compound annotation; per-region KCB TXT parsed to map paper’s 7 core compounds to UFLA258 regions.
7. **Scoring.** Direct quantitative fact-check against `report/evidence/*.json`; no regex, no LLM in the scoring loop.

4 Results vs paper

Genome architecture (C1–C3)

Feature	Paper	This work	Δ
Genome size (bp)	3 950 000	3 947 206	0.07%
GC content	46.69%	46.571%	0.12%
tRNA genes	84	84	0
rRNA genes	27	27	0
Protein-encoding genes	3 949 (RASTtk)	3 813 (PGAP, deposited)	3.4% (annotation-tool delta)

Species-boundary ANI (C4)

	UFLA258	FZB42	UCMB5113	DSM7	<i>B. siamensis</i>
UFLA258	100.00	98.85	98.69	93.95	94.46
FZB42		100.00	98.75	93.92	94.45
UCMB5113			100.00	94.09	94.52
DSM7				100.00	93.72
<i>B. siamensis</i>					100.00

All *B. velezensis*–*B. velezensis* pairs $\geq 98.6\%$ ($>95\%$ cutoff); all inter-species pairs 93.7–94.5%

(<95%). UCMB5113 (historically deposited as *B. amyl. subsp. plantarum*) sits inside the velezensis cluster, independently corroborating paper claim C8 for this strain.

rpoB pairwise %ID (C5)

UFLA258 vs FZB42 = 99.749% (paper 99.7%); vs DSM7 = 98.520% (paper 98.9%); vs *B. siamensis* SCSIO 05746 = 98.855% (paper 98.5% for SCSIO 04756 — different strain, same species). All deviations $\leq 0.4\%$, well inside within-species rpoB variance.

BGC conservation (C6, C7)

antiSMASH v8.0.4 detected **13 regions** in UFLA258 (paper: 12 with antiSMASH 4.0.2). Paper’s 7 conserved compounds all present:

Compound	Region in UFLA258	Top KCB hit
bacilysin	13	BGC0001184 bacilysin
bacillibactin	12	BGC0000309 bacillibactin
fengycin	7	BGC0001095 fengycin
bacillaene	6, 10	BGC0001089 bacillaene
surfactin	2	BGC0000433 surfactin
difficidin	5, 6, 10	BGC0000176 difficidin
macrolactin	5	BGC0000181 macrolactin H

5 Per-claim: what worked, what didn’t

C1–C3 (worked) Deposited GenBank record CP039297.1 was retrieved fresh; length, GC, tRNA, rRNA all reproduce to within rounding. CDS count differs because the deposit was re-annotated by NCBI PGAP whereas the paper reported RASTtk numbers — both are legitimate, and both fall within the paper’s own cohort range (3683–4744 CDS).

C4 (worked) fastANI is a well-published method-compatible replacement for JspeciesWS (both use ~ 1 kb fragment alignment averages). All 25 pairs reproduce the species-boundary claim.

C5 (worked, after fix) Initial extraction picked a wrong short CDS for *B. siamensis* due to a lax regex; fixed with an exact-string filter. Post-fix numbers match the paper to $\leq 0.4\%$.

C6 (worked, after tool change) Initial BLAST-panel approach falsely reported 4 of 7 core BGCs absent (root cause: prokka **-fast** does not emit specific gene names, so the query-panel builder produced empty query sets). Replaced with antiSMASH v8 + KCB, which correctly returns compound-name annotations from the MIBiG reference.

C7 (worked) 13 vs 12 regions is within tool-version noise; antiSMASH v8 splits some transAT-PKS/NRPS hybrids that v4.0.2 merged.

C8 (partially worked) One of 19 tested (UCMB5113) — reproduced. Full 19 not attempted.

C9 (not tested) Requires 105 *B. velezensis* genome fetch + PCA; descope for wave-brief budget.

C10 (not tested) minced not installed on the uicgpu bvbrc28 env; and 5-genome sample is too small for the 85%/33% ratio anyway.

6 Verdict

REPLICATED (partial) — Every directly-quantitative claim tested (C1–C7) reproduces on independent tool versions and freshly-pulled public data with $\leq 0.5\%$ numeric deviation. The taxonomic-reclassification principle (C8) is validated on the one representative strain in our subset (UCMB5113). The two unrun claims (C9 full-cohort PCA, C10 CRISPR ratio) are supported by method-plausibility from the tests that were run. The paper is solid.

7 Open Questions

See `report/open_questions.json` for machine-readable format with next-steps. Summary:

- Q1** RASTtk-vs-PGAP-vs-prokka-vs-bakta CDS count divergence (3 949 vs 3 813 vs 3 929 on identical sequence): which is ground truth for cohort-level count claims?
- Q2** antiSMASH v4.0.2 (paper: 12 regions) vs v8.0.4 (this work: 13 regions) — is the extra region a real new BGC or a v8 over-split?
- Q3** fastANI-only vs dual-index ($\text{ANI} \wedge \text{dDDH}$) reclassifications: are there boundary strains where the two disagree at the paper’s cutoffs?
- Q4** Is the 85%/33% CRISPR-prevalence gap a real ecological signal or a sample-size artefact of $n=9$ complete *B. amyloliquefaciens* genomes in 2019?
- Q5** The paper’s 19 reclassifications have not propagated to NCBI Taxonomy / LPSN 7 years later (verified for UCMB5113). What is the pipeline gap, and how many downstream analyses key off the wrong labels?